STUDY COACH: R2. BASELINE AND PROPOSAL

Dipti Aswath **Buddhika Makehelwala** **Jiashan Cui** **Khubi Shah**
{daswath, bmakehelwala, jiashanc, kshah}@andrew.cmu.edu

1 BASELINE MODELS AND METRICS (2 PAGES)

We address **incongruence detection**: detecting mismatches between student explanations and scientific figures. Starting from SPIQA Test-A Pramanick et al. (2024)-a manually curated split of 666 QA pairs across 118 papers-we constructed **SPIQA+**, an augmented evaluation dataset, via a three-phase pipeline:

1. **Prepare:** Filter SPIQA Test-A to figure-grounded QAs and annotate figure types (table, plot, figure/schematic).
2. **Generate:** Hand-curate 18 In-Context Learning (ICL) exemplars covering all combinations: 3 figure types $\times$ 3 error categories (factual, conceptual, omission) $\times$ 2 verdicts (incorrect, partially correct). Use GPT-5.1 to synthetically generate incorrect and partially correct student answers with structured model responses.
3. **Validate:** Human spot-check ($\sim$20 examples), analyze error distribution, and enforce a quality gate ($\geq$90% agreement).

Each generated example follows a standardized **4-tuple** format: (1) *Context*-visual + textual context from the original SPIQA paper (figure + caption); (2) *Question*-the figure-grounded question from SPIQA; (3) *User Answer*-a synthetically generated student explanation (correct, partially correct, or incorrect); and (4) *Model Response*-a structured ground-truth containing a verdict, error category, and free-text coaching feedback (2–4 sentences). The resulting SPIQA+ dataset contains **174 generated examples** with the following distributions: error types are 48% factual, 38% conceptual, and 14% omission; figure types are 70 tables, 68 plots, 32 schematics, and 4 other.

Task setup. Given a question about a scientific figure and a student's answer - which may be correct, partially correct, or incorrect - the model must:

1. *Detect mismatch* - classify the student's answer as correct, partially correct, or incorrect.
   "Is this explanation right?"
2. *Categorize the error type* - identify whether the mistake is factual, conceptual, or an omission.
   "What kind of mistake?"
3. *Provide coaching feedback* - localize and explain the error with constructive guidance.
   "Here's what you missed..."

This task is fundamentally multimodal: verifying "BERT achieves 89%" requires reading the actual bar height (84.6%) from the figure (visual grounding), while detecting a wrong conclusion requires mapping textual interpretations onto visual patterns (cross-modal reasoning).

The SPIQA paper established that paper context improves figure QA performance (Gemini 1.5 Pro achieves 8+ point L3Score improvements with full context). However, SPIQA only addresses: *Can models generate correct answers?* It does not address: *Can models evaluate whether a human's explanation is correct?* This answer evaluation capability is essential for educational applications like interview coaching. Because our model is evaluating a student's answer rather than generating its own, it receives **no reference answer** - it must reason directly from the figure and context to judge correctness. Including reference answers in early experiments reduced modality differences and lowered alignment with human judgment, so we removed them.

Experimental conditions. We test four input conditions to ablate the contribution of each modality:

- **C1 (Text-Only):** Question + student answer only.
  No figure, no caption.
- **C2 (Caption-Only):** C1 + figure caption.
  Textual description of the figure, but no image.
- **C3 (Vision-Only):** C1 + figure image.
  Raw visual input, but no caption.
- **C4 (Multimodal):** C1 + figure image + caption.
  Full context available to the model.

Hypotheses. We test four hypotheses across these conditions:

1. **H1 (Multimodal Advantage):** Visual input improves performance over text-only.
2. **H2 (Metric Alignment):** Verdict accuracy and feedback quality will be correlated.
3. **H3 (Error Type):** Factual errors will be detected more reliably than conceptual errors, and visual context will help conceptual errors more.
4. **H4 (Figure Type):** Tables will be easiest (structured, localized information) and schematics hardest (distributed spatial relationships).

1.1 Unimodal Baselines

Our unimodal baselines use **Qwen3-VL-8B** Bai et al. (2023) under conditions C1 and C2, which provide only textual information. The Qwen-VL architecture uses a Vision Transformer (ViT) visual encoder initialized from OpenCLIP Bai et al. (2023), with a position-aware cross-attention adapter that compresses visual features to a fixed-length sequence of 256 tokens, which is then fed to the LLM backbone. This architecture was trained via a three-stage pipeline: large-scale image-text pretraining, multi-task pretraining at higher resolution (448×448), and supervised fine-tuning for instruction following.

C1 (Text-Only) provides only the question and student answer. The model must rely on pre-trained knowledge to judge whether a claim like "BERT achieves 89% accuracy" sounds plausible. C1 achieves the *highest* verdict accuracy at 8B (56%), revealing that the model does **plausibility checking**-judging whether claims sound reasonable-rather than verifying claims against the figure.

C2 (Caption-Only) adds the figure caption without the image. At 8B, C2 underperforms C1 on verdict (48% vs 56%), suggesting that at small scale, the caption introduces noise rather than helping classification. However, C2 shows improved feedback quality (46% soft match vs 40% for C1).

1.2 Simple Multimodal Baselines

We extend to multimodal conditions using the same Qwen3-VL-8B model under C3 and C4, which add figure images:

C3 (Vision-Only) provides the figure image but no caption. At 8B, C3 achieves 50% verdict accuracy and 48% soft match-better than C2 on feedback but worse than C1 on verdict.

C4 (Multimodal) provides both figure image and caption, giving the model full context. At 8B, C4 achieves the worst verdict accuracy (48%) but the best feedback quality: 6% exact match and 56% soft match. Human evaluation on n=10 per condition confirmed this pattern even more dramatically: C4 achieved 80% human-evaluated match versus C1's 40% (+40pp). This dissociation-*the model classifies worse but explains better with images*-was the central surprising finding from our midterm evaluation.

1.3 Competitive Approaches

To test whether model scale resolves the verdict–feedback dissociation, we evaluate **Qwen3-VL-32B** Bai et al. (2023) across all four conditions. For direct comparison with the 8B baseline, Table 1 reports N=50 per condition. Both models were also run on the full N=174 sample for error type

(H3, Table 2) and figure type (H4, Table 3) analyses. We also evaluated LLaVA-1.5-7B Liu et al. (2023) on the error type hypothesis; Qwen3-VL-32B outperformed it and was the only model where visual context improved conceptual verdict accuracy.

1. **Qwen3-VL-32B (C4, Multimodal):** Our primary competitive approach. At 32B, C4 achieves 20% exact feedback match - a $10\times$ improvement over C1's 2% and a $3.3\times$ improvement over the 8B C4 result (6%). The verdict gap narrows from -8pp (8B) to -2pp (32B), suggesting scaling enables the model to extract useful signal from visual input.

2. **Qwen3-VL-32B (C2, Caption-Only):** Achieves the best verdict accuracy at 32B (58%), demonstrating that at scale, textual figure descriptions provide the most reliable classification signal. The $+10$pp improvement over 8B C2 (48%) is the largest single-condition gain from scaling.

3. **Qwen3-VL-32B (C1, Text-Only):** Serves as the scaled text-only reference. C1 verdict accuracy is unchanged at 56%, but feedback quality remains poor (2% match) - confirming that text-only feedback does not improve with scale.

Table 1: H1/H2: Verdict accuracy and feedback quality by condition (N=50 per condition). Verdict = three-class classification (correct/partially correct/incorrect). Match % and Soft Match % = LLM-as-Judge (Claude Sonnet 4, temp=0). Human Match = manual evaluation (n=10, 8B only). H1 is read vertically (C1 vs C4); H2 is read horizontally (verdict vs feedback).

	Qwen3-VL-8B				Qwen3-VL-32B			
Condition	Verdict	Match	Soft M.	Human	Verdict	Match	Soft M.	Human
C1: Text-Only	**56%**	2%	40%	40%	56%	2%	42%	-
C2: Caption-Only	48%	2%	46%	50%	**58%**	8%	48%	-
C3: Vision-Only	50%	4%	48%	60%	56%	6%	48%	-
C4: Multimodal	48%	6%	56%	80%	54%	**20%**	**58%**	-
Δ (C4−C1)	−8pp	+4pp	+16pp	+40pp	−2pp	+18pp	+16pp	-

Table 2: H3: Verdict accuracy and feedback match by error type (N=174; 52 factual, 41 conceptual, 15 omission). Avg = average across C1–C4. Δ = C4−C1, measuring the benefit of visual context.

	Qwen3-VL-8B				Qwen3-VL-32B			
Error Type	Verd. Avg	Verd. Δ	Fdbk Avg	Fdbk Δ	Verd. Avg	Verd. Δ	Fdbk Avg	Fdbk Δ
Factual	39.4%	−21.2pp	25.5%	+21.2pp	28.4%	−7.7pp	26.0%	+21.2pp
Conceptual	54.3%	−14.6pp	43.3%	+31.7pp	57.3%	**+2.4pp**	42.1%	+29.3pp
Omission	45.0%	−6.7pp	15.0%	+33.3pp	48.3%	−26.7pp	13.4%	+26.7pp

2 RESULTS (1 PAGE)

Tables 1–3 summarize results across all four hypotheses, both model scales, and four input conditions. We evaluate three complementary metrics.

Metric 1: Verdict Accuracy (three-class classification). The model classifies the student's answer as correct, partially correct, or incorrect. This is the primary extrinsic metric for StudyCoach's grading component. **H1** (Table 1): At 8B, text-only C1 achieves the best verdict (56%) while multimodal C4 is worst (48%, Δ=−8pp). At 32B, the gap narrows: C2 leads at 58%, C4 reaches 54% (Δ=−2pp). **H3** (Table 2): Conceptual errors are easier than factual - the opposite of our hypothesis. Factual accuracy is just 28.4% at 32B; conceptual reaches 57.3%. At 32B, conceptual is the *only* error type where visual context helps verdict (+2.4pp). **H4** (Table 3): Schematics benefit most from visual context at both scales (+9.4pp at 8B, +18.8pp at 32B), while plots are consistently hurt (−10.3pp at 8B, −4.4pp at 32B).

Metric 2: Feedback Match Rate (LLM-as-Judge). We use Claude Sonnet 4 as an automated judge to classify each response as Match, Partial, or Unmatched against ground-truth explanations. **H1**: C4 at 32B achieves 20% exact match - a 10× improvement over C1's 2%. Visual context helps feedback for *all* error types (+21–33pp, Table 2) and *all* figure types (+7–34pp, Table 3) at both scales. **H2**: Verdict accuracy and feedback quality are *inversely correlated* - C1/C2 classify best but produce the worst feedback, while C4 classifies worst but explains best. Traditional auto metrics (F1: 0.29–0.33, ROUGE-L: 0.18–0.21, BLEU: 4.8–8.1) fail to differentiate between conditions at either scale; only LLM-as-Judge and human evaluation capture meaningful differences.

Metric 3: Soft Match Rate (Match + Partial). Soft Match counts feedback that is at least partially useful for coaching. C4 at 32B reaches 58% vs 42% for C1. Human evaluation on a subset (n=10 per condition, 8B only) confirmed the pattern: human match ranged from 40% (C1) to 80% (C4), validating the LLM-as-Judge methodology.

Key insight: The verdict–feedback dissociation (H2) holds across all error types and figure types at both scales. StudyCoach should use condition routing: text-only for grading, multimodal for coaching.

Table 3: H4: Verdict accuracy and feedback match by figure type (N=174; 70 tables, 68 plots, 32 schematics). Avg = average across C1–C4. Δ = C4−C1.

	Qwen3-VL-8B				Qwen3-VL-32B			
Figure Type	Verd. Avg	Verd. Δ	Fdbk Avg	Fdbk Δ	Verd. Avg	Verd. Δ	Fdbk Avg	Fdbk Δ
Table	51.4%	−2.9pp	6.4%	+7.1pp	51.1%	+1.4pp	12.1%	+15.7pp
Plot	47.8%	−10.3pp	21.7%	+19.1pp	49.3%	−4.4pp	26.5%	+33.8pp
Schematic	48.4%	+9.4pp	35.2%	+25.0pp	**57.0%**	**+18.8pp**	**46.1%**	+25.0pp

3 MODEL PROPOSAL (1 PAGE)

Based on our empirical findings across four hypotheses, two model scales, and four input conditions, we propose a **condition-routing architecture** for StudyCoach.

3.1 OVERALL MODEL STRUCTURE

The central finding is that no single condition optimizes both verdict accuracy and feedback quality. We therefore propose a **condition-routing architecture** with three levels of routing informed by our empirical results.

Task mode routing. The verdict–feedback dissociation (H2) motivates separate routing by task:

Task	Recommended Condition	Rationale
Grading (verdict)	C1 · Text-Only (8B) or C2 · Caption-Only (32B)	Visual input hurts classification at both scales; caption helps at 32B (+2pp over C1)
Coaching (feedback)	C4 · Multimodal at 32B	Best feedback quality by a wide margin (20% match vs 2–8% others); model uses the figure to explain
Combined	C4 · Multimodal at 32B	Small verdict cost (-2pp) is outweighed by $10\times$ feedback improvement (2%→20%)

Table 4: Task mode routing based on 8B vs 32B results.

Error type routing (from H3, N=174). Different error types require different input strategies:

Error Type	Best Condition	Verd. Δ	Fdbk Δ	Rationale
Factual	8B or 32B · C1 (Text-Only)	-7.7pp	+21.2pp	Visual context hurts verdict at both scales; model cannot reliably ground specific values. CoT or 72B+ needed.
Conceptual	32B · C4 (Multimodal)	+2.4pp	+29.3pp	Only model + condition where visual context helps verdict accuracy. Strong feedback gains.
Omission	C1 for verdict; C4 for feedback	-26.7pp	+26.7pp	Visual context severely hurts verdict (worse at 32B than 8B) but helps feedback. Split strategy recommended.

Table 5: Error type routing. Δ = C4$-$C1, measuring visual context benefit.

Figure type routing (from H4, N=174). Visual context effectiveness varies dramatically by figure type:

Figure Type	8B Verd.	32B Verd.	Recommendation
Tables	51.4%	51.1%	Similar accuracy at both scales ($\sim$51%). Visual context hurts slightly at 8B (-2.9pp), helps slightly at 32B (+1.4pp).
Plots	47.8%	49.3%	Visual context hurts verdict at both scales. Consider CoT prompting or 72B+ for trend verification.
Schematics	48.4%	57.0%	32B with visual context best (+18.8pp); 8B also benefits (+9.4pp). Essential for feedback quality at both scales.

Table 6: Figure type routing. Verdict accuracy averaged across C1–C4.

Sections 3.2–3.4 detail the architectural components and training objectives that implement this routing, comparing the LLaVA-1.5 cross-modal transfer approach with Qwen3-VL's co-learning approach and identifying where MoE scaling may address remaining challenges.

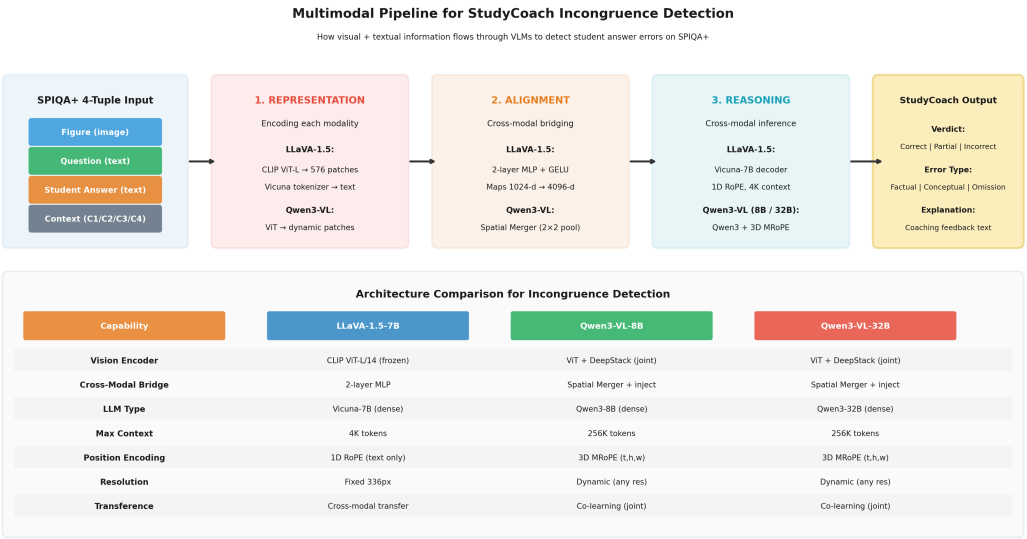

Figure 1: Multimodal pipeline for StudyCoach: SPIQA+ 4-tuple input → representation (encoding each modality) → alignment (cross-modal bridging) → reasoning & generation (cross-modal inference) → coaching output. The architecture comparison table shows how LLaVA-1.5-7B, Qwen3-VL-8B, and Qwen3-VL-32B differ across each stage.

3.2 Encoders

Our models represent two distinct approaches to the multimodal *transference* challenge Liang et al. (2024) - transferring knowledge between visual and textual modalities.

LLaVA-1.5-7B: Cross-modal transfer (frozen encoder). LLaVA-1.5 Liu et al. (2023) uses CLIP ViT-L/14 at fixed 336×336 resolution (576 patches × 1024-d, penultimate layer features). The encoder is completely frozen - a pure cross-modal transfer approach where CLIP's web-scale contrastive features are projected into the LLM's space without adaptation. This limits scientific figure understanding to what CLIP learned from general image-text pairs, which may not support the fine-grained numerical extraction our task demands (e.g., distinguishing a bar at 84.6% from one at 89%).

Qwen3-VL: Co-learning (jointly trained encoder). Qwen3-VL Bai et al. (2023) uses a ViT with dynamic resolution (variable patch counts preserving native aspect ratio) and 2D-RoPE encoding spatial row-column position within the ViT itself. Crucially, this encoder is *jointly trained* with the LLM during pre-training - a co-learning approach where visual and linguistic representations co-adapt. This directly addresses transference: rather than transferring frozen features across a modality gap, the encoder learns to produce features the decoder can effectively use. Our H4 results confirm that the visual heterogeneity of SPIQA+ (tables 44.3%, plots 28.1%, schematics 17.4%) creates different encoding challenges - tables are easiest for verdict (51.4% avg) but hardest for feedback (6.4% match), while schematics benefit most from visual context (+9.4pp verdict, +25.0pp feedback at 8B).

Alternative: Caption-as-encoder. Our C2 results show that figure captions alone achieve the best verdict accuracy at 32B (58%). A lightweight alternative would encode figures as structured text (caption + OCR) rather than raw pixels, entirely bypassing the visual encoder. This is especially promising for tables, where spatial cell relationships are difficult for vision models.

3.3 Decoders

LLaVA-1.5-7B: Shallow alignment. Uses Vicuna-7B v1.5 (32 layers, 4096-d, 1D RoPE, ∼4K context). Visual tokens from a 2-layer MLP projector (GELU, 1024→4096) are concatenated with

text tokens before the first layer - a shallow cross-modal bridge that creates a large modality gap between visual and linguistic representations.

Qwen3-VL: Deep multimodal injection. Uses a 36-layer Qwen3 decoder with Interleaved-MRoPE encoding three spatial dimensions (temporal, height, width), enabling the decoder to ground spatial claims to specific figure regions. Its DeepStack mechanism injects multi-level ViT features from layers 8, 16, and 24 into early LLM layers, creating multiple transference pathways that preserve low-level visual details (character shapes for OCR, edges for trend detection). This deep injection reduces the modality gap that limits LLaVA's approach, and the 256K context window accommodates full paper context for a future C5 condition.

Our 32B scaling results support the co-learning advantage: multimodal feedback match improved from 6% (8B) to 20% (32B), and schematics' visual context benefit doubled from +9.4pp to +18.8pp for verdict accuracy - gains consistent with deeper visual-linguistic integration at scale.

Scaling path: Mixture-of-Experts (MoE). Qwen3-VL-30B-A3B replaces standard FFN layers with MoE modules (128 experts, 8 active per token, $\sim$3B active parameters of 30B total). This adds a transference dimension: different experts may specialize in visual-token processing vs language reasoning, creating implicit modality-specialized sub-networks. A global-batch load balancing loss prevents expert collapse and encourages specialization across our error categories. The MoE variant provides quality approaching 32B dense at $\sim$8B compute cost - and the larger Qwen3-VL-235B-A22B (22B active) represents our primary scaling target. Our 8B$\rightarrow$32B results showed that scale specifically helps where visual grounding matters most: conceptual verdict flipped from -14.6pp to +2.4pp, and schematic benefit doubled from +9.4pp to +18.8pp. MoE scaling may extend these gains to factual errors (currently 28.4% avg) and plots (-4.4pp) that remain challenging at 32B.

Alternative: Chain-of-thought visual decoding. Force the model to *describe* the figure before judging the student's answer. This makes visual attention explicit and may close the verdict gap for plots (-4.4pp at 32B) and factual errors (-7.7pp) where the model currently fails to ground specific values.

3.4 Loss Functions

Primary: Auto-regressive cross-entropy. All models use $\mathcal{L} = -\sum \log P(t_i \mid V, t_1 \dots t_{i-1})$ over response tokens only. Qwen3-VL additionally uses *square-root reweighting*, scaling multimodal sample loss by the square root of the text-to-multimodal data ratio during pre-training. This prevents degradation on pure-text understanding while learning visual grounding - directly relevant since our task requires simultaneously parsing textual claims and verifying them against figures. Our results confirm this dual capability matters: visual context helps *all* error types for feedback (+21.2 to +33.3pp) even while hurting verdict accuracy.

Auxiliary 1: Verdict classification head. A separate cross-entropy loss on a classification head extracting the hidden state at the verdict token position, supervised with 3-class labels. This targets the verdict–feedback dissociation (H2): optimizing feedback generation actually degrades classification, so a dedicated head provides direct supervisory signal for the detection task.

Auxiliary 2: Contrastive visual grounding loss. LLaVA's encoder was shaped by CLIP's contrastive loss (InfoNCE), which provides coarse-grained semantic matching but not the precise numerical extraction needed to verify specific claims. This may explain why visual input hurts verdict for factual errors (-21.2pp at 8B, -7.7pp at 32B) - the correct value exists only in the figure, the claimed value only in text, and the model must extract and compare both precisely. A fine-grained contrastive loss on (claim, figure region) pairs would target this weakness.

Auxiliary 3: Feedback–reference alignment loss. A soft alignment loss between generated feedback and ground-truth explanation embeddings, targeting the low match rates (2–20%) by providing gradient signal toward more precise explanations.

Auxiliary 4: MoE load balancing loss (for MoE scaling). When fine-tuning the MoE variant, a global-batch load balancing loss ensures well-distributed expert routing, enabling specialized circuits for different sub-tasks - visual grounding for factual errors, spatial reasoning for schematics, and textual analysis for conceptual errors.

3.5 TEAM MEMBER CONTRIBUTIONS TO THIS REPORT

Dipti Aswath constructed 9 of the 18 ICL exemplars for SPIQA+ and led dataset validation (Phase 3: human spot-check of $\sim$20 examples, error distribution analysis, quality gate enforcement at $\geq$90% agreement). Ran all H3 (error type) and H4 (figure type) experiments with the baseline model (Qwen3-VL-8B, N=174), producing both verdict accuracy and feedback quality results. Designed the hypothesis testing framework, performed cross-hypothesis analysis and synthesis (including the verdict–feedback dissociation finding and condition-routing proposal), and wrote Report 2.

Buddhika Makehelwala led SPIQA+ dataset generation (Phase 2: GPT-5.1 synthetic generation of incorrect and partially correct student answers using the 18 ICL exemplars, producing the 174-example dataset). Selected the competitive model (Qwen3-VL-32B) and ran all competitive model experiments: H1/H2 (N=50) and H3/H4 (N=174) across all four conditions.

Jiashan Cui developed and ran the evaluation pipeline for H1/H2 with the baseline model (Qwen3-VL-8B): automated metrics (F1, ROUGE-L, BLEU), LLM-as-Judge evaluation (Claude Sonnet 4), and human annotation (n=10 per condition). Extended evaluation to the competitive model (Qwen3-VL-32B) for H1/H2 with human evaluation and LLM-as-Judge, and ran human evaluation for H3/H4 competitive model results.

Khubi Shah ran the baseline model (Qwen3-VL-8B) for H1/H2 across all four conditions (N=50). Authored the model proposal components: Encoders (Section 3.2), Decoders (Section 3.3), and Loss Functions (Section 3.4).

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A Appendix

A.1 Hypothesis Testing Summary

We tested four hypotheses across two model scales (Qwen3-VL-8B and Qwen3-VL-32B):

Hypothesis	Prediction	Verdict Result	Feedback Result
H1: Multimodal Advantage	Visual input helps	FAIL (-8pp at 8B, -2pp at 32B)	PASS ($+16$pp soft match, 8B)
H2: Metric Alignment	Verdict $\propto$ feedback	FAIL (inversely correlated)	LLM Judge + Human validate
H3: Error Type	Factual $>$ Conceptual	FAIL (28.4% $<$ 57.3% at 32B)	Context helps conceptual more
H4: Figure Type	Tables $>$ Schematics	MIXED (32B: 51.1% $<$ 57.0%)	Schematics best at both scales

Table 7: Hypothesis results summary across both model scales.

A.2 Error Type Analysis (H3 Detail)

Model	Error Type	C1	C2	C3	C4	Avg	Δ (C4$-$C1)
8B	Factual (n=52)	51.9%	40.4%	34.6%	30.8%	39.4%	-21.2pp
	Conceptual (n=41)	63.4%	61.0%	43.9%	48.8%	54.3%	-14.6pp
	Omission (n=15)	53.3%	33.3%	46.7%	46.7%	45.0%	-6.7pp
32B	Factual (n=52)	32.7%	28.8%	26.9%	25.0%	28.4%	-7.7pp
	Conceptual (n=41)	58.5%	56.1%	53.7%	61.0%	57.3%	**+2.4pp**
	Omission (n=15)	66.7%	53.3%	33.3%	40.0%	48.3%	-26.7pp

Table 8: Verdict accuracy by error type and condition. Conceptual at 32B C4 is the *only* cell where visual context improves verdict accuracy.

Model	Error Type	C1	C2	C3	C4	Avg	Δ (C4$-$C1)
8B	Factual	13.5%	25.0%	28.8%	34.6%	25.5%	+21.2pp
	Conceptual	26.8%	51.2%	36.6%	58.5%	43.3%	+31.7pp
	Omission	0.0%	0.0%	26.7%	33.3%	15.0%	+33.3pp
32B	Factual	13.5%	26.9%	28.8%	34.6%	26.0%	+21.2pp
	Conceptual	26.8%	48.8%	36.6%	56.1%	42.1%	+29.3pp
	Omission	0.0%	0.0%	26.7%	26.7%	13.4%	+26.7pp

Table 9: Feedback match rate (LLM-as-Judge) by error type and condition. Visual context helps feedback quality for *all* error types at both scales ($+19$–33pp).

A.3 Figure Type Analysis (H4 Detail)

A.4 8B $\rightarrow$ 32B Scaling Comparison

Key improvements at 32B: (1) Multimodal feedback quality jumps from 6% to 20% match ($+14$pp). (2) Schematics: visual context benefit doubles for verdict ($+18.8$pp vs $+9.4$pp at 8B). (3) Conceptual errors: visual context now *helps* verdict ($+2.4$pp vs -14.6pp at 8B). (4) Plot feedback: context benefit nearly doubles ($+33.8$pp vs $+19.1$pp).

Model	Figure Type	C1	C2	C3	C4	Avg	Δ (C4−C1)
8B	Table (n=70)	52.9%	51.4%	51.4%	50.0%	51.4%	−2.9pp
	Plot (n=68)	55.9%	47.1%	42.6%	45.6%	47.8%	−10.3pp
	Schematic (n=32)	50.0%	46.9%	37.5%	59.4%	48.4%	**+9.4pp**
32B	Table (n=70)	51.4%	51.4%	48.6%	52.9%	51.1%	+1.4pp
	Plot (n=68)	52.9%	47.1%	48.5%	48.5%	49.3%	−4.4pp
	Schematic (n=32)	46.9%	56.2%	59.4%	65.6%	**57.0%**	**+18.8pp**

Table 10: Verdict accuracy by figure type and condition (N=174). Schematics benefit most from visual context at both scales; plots are consistently hurt.

Model	Figure Type	C1	C2	C3	C4	Avg	Δ (C4−C1)
8B	Table (n=70)	1.4%	4.3%	11.4%	8.6%	6.4%	+7.1pp
	Plot (n=68)	13.2%	25.0%	16.2%	32.4%	21.7%	+19.1pp
	Schematic (n=32)	25.0%	37.5%	28.1%	50.0%	**35.2%**	+25.0pp
32B	Table (n=70)	4.3%	7.1%	17.1%	20.0%	12.1%	+15.7pp
	Plot (n=68)	10.3%	32.4%	19.1%	44.1%	26.5%	+33.8pp
	Schematic (n=32)	31.2%	50.0%	46.9%	56.2%	**46.1%**	+25.0pp

Table 11: Feedback match rate (LLM-as-Judge) by figure type and condition. Schematics achieve the highest feedback quality at both scales; tables are worst. Visual context helps feedback for *all* figure types at both scales.

Still challenging at 32B: (1) Factual errors remain hard (28.4% avg) with visual context still hurting verdict (−7.7pp). (2) Text-only feedback quality unchanged (2% match at both scales). (3) Plots still hurt by visual context for verdict (−4.4pp). (4) Omission errors: visual context hurts verdict *more* at 32B (−26.7pp vs −6.7pp at 8B).

A.5 Auto Metrics vs. LLM-as-Judge

Traditional auto metrics fail to capture feedback quality differences across conditions:

Metric	8B Range	32B Range	Differentiates?
F1	0.29 – 0.30	0.294 – 0.325	No
ROUGE-L	0.18 – 0.20	0.192 – 0.207	No
BLEU	4.8 – 6.7	5.93 – 8.13	No
LLM Judge (Soft Match)	40% – 56%	42% – 58%	Yes
Human Eval (Match)	40% – 80%	-	Yes

Table 12: Evaluation method comparison. Only LLM-as-Judge and human evaluation differentiate feedback quality across conditions.

A.6 Statistical Notes

With n=50 (H1/H2 testing), differences <5pp are noise (∼7pp confidence intervals). The 8pp gap between text-only (56%) and multimodal (48%) verdict accuracy at 8B is borderline significant. Human evaluation used n=10 per condition-each example shifts match rate by 10pp. H3 and H4 use the full N=174 sample for both models, providing more reliable estimates.

A.7 Next Steps

Priority actions: (1) Chain-of-thought prompting to force explicit figure description before judgment-targets factual errors and plot interpretation; (2) Test Qwen3-VL-235B-A22B (MoE, 22B active parameters)-the most powerful Qwen VL model may extend scaling gains to factual errors; (3) Larger evaluation samples (200+) to shrink confidence intervals from ∼7pp to ∼3–4pp; (4) Add

C5 condition (multimodal + paper paragraphs) for conceptual and omission errors that require understanding experimental setup beyond the figure.